

EG1311: Design & Make

Driver & Sensor

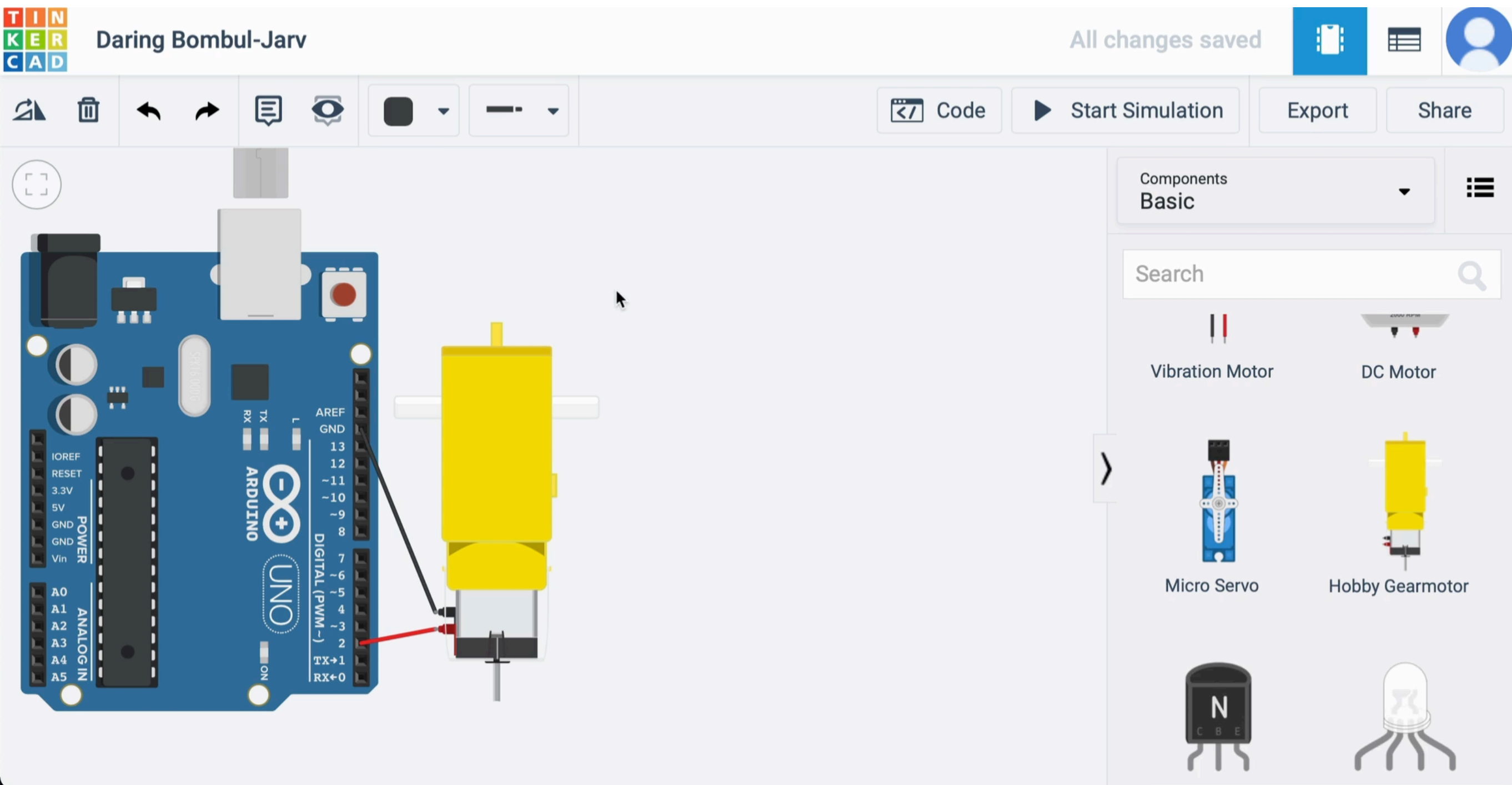
Dr. Jason S. Ku

Motor Demo

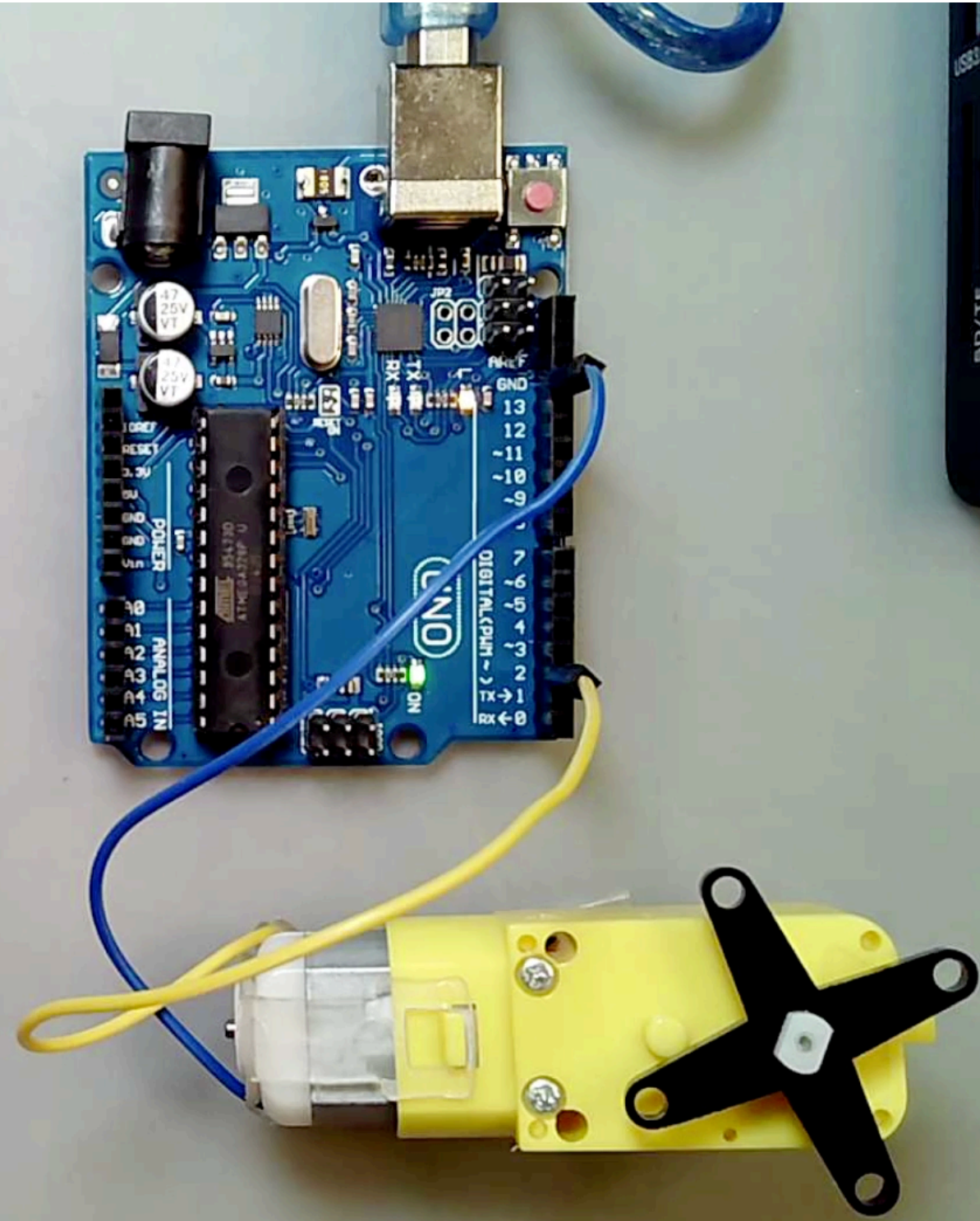
```
void setup() {  
    pinMode(2, OUTPUT);  
    digitalWrite(2, HIGH);  
}
```

```
void loop() {  
}
```

Motor Demo

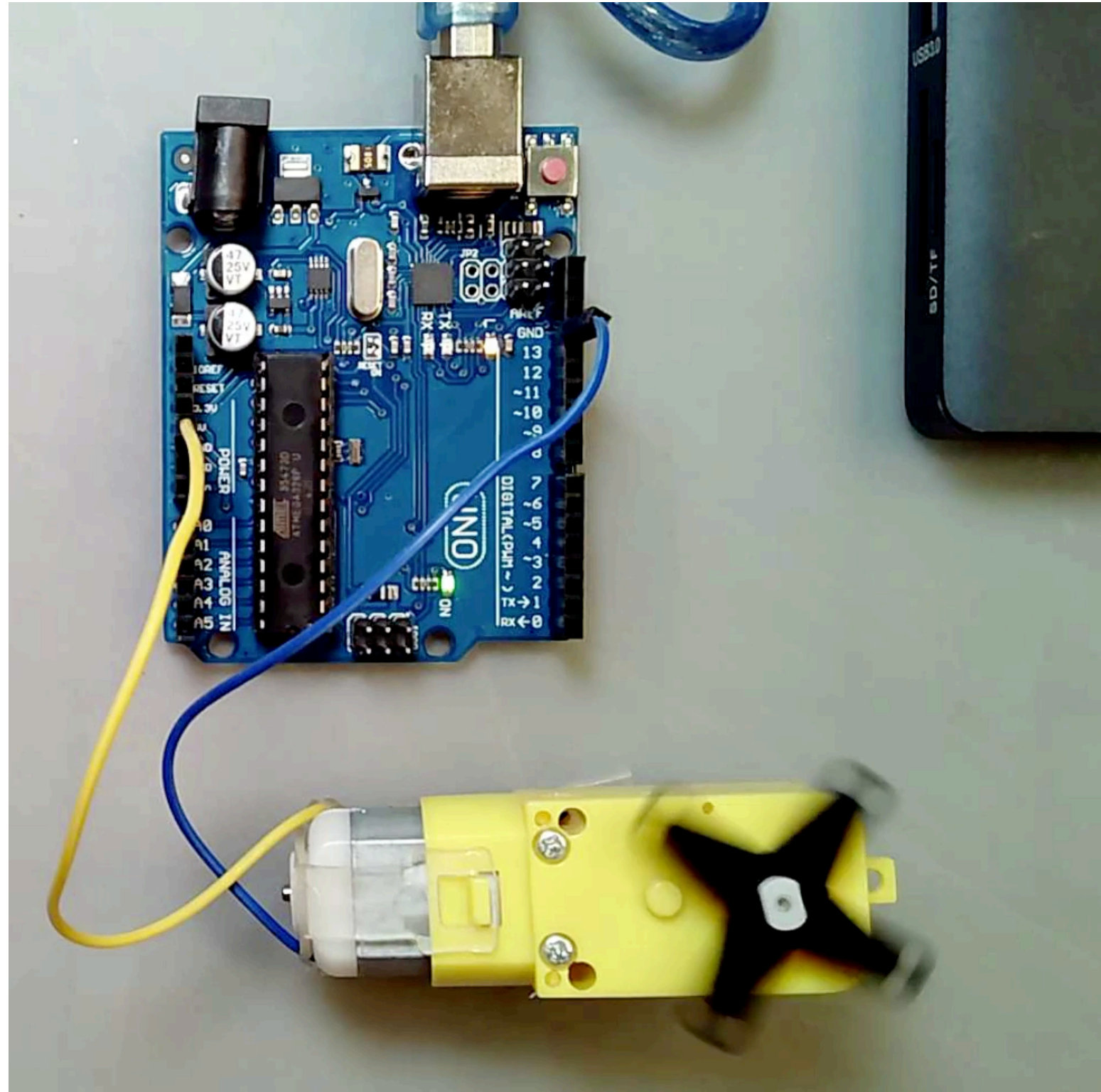
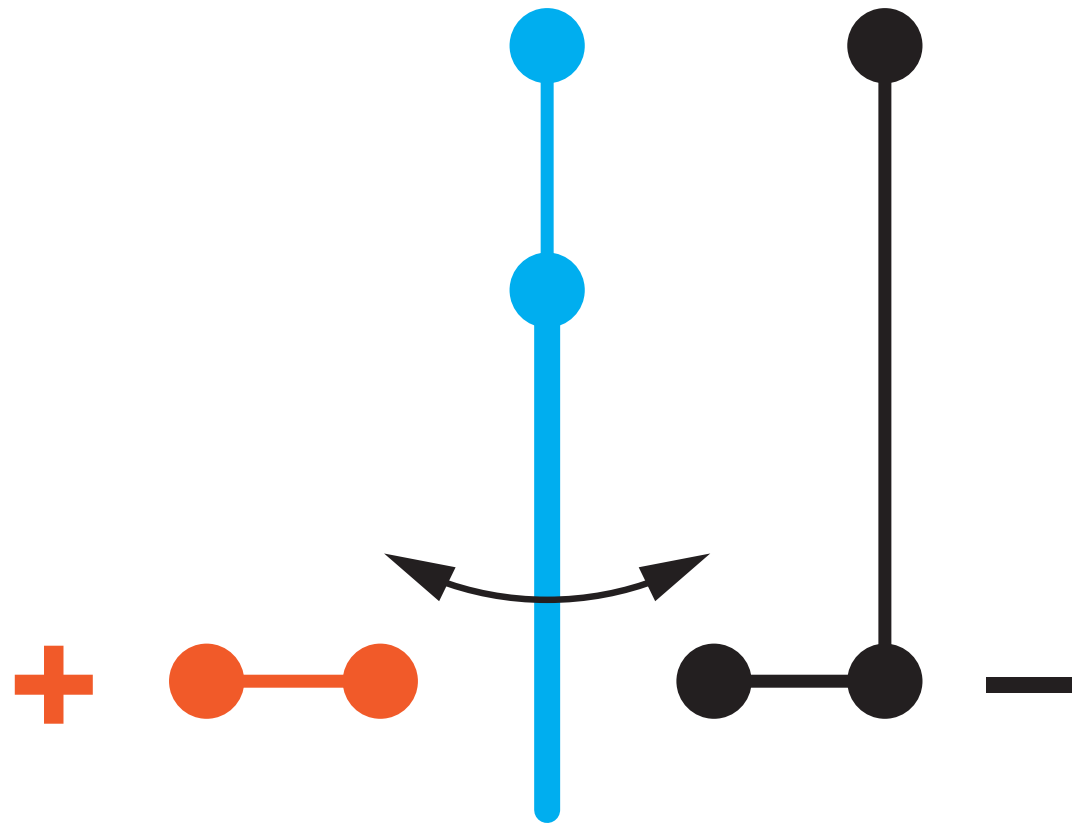


Motor Demo

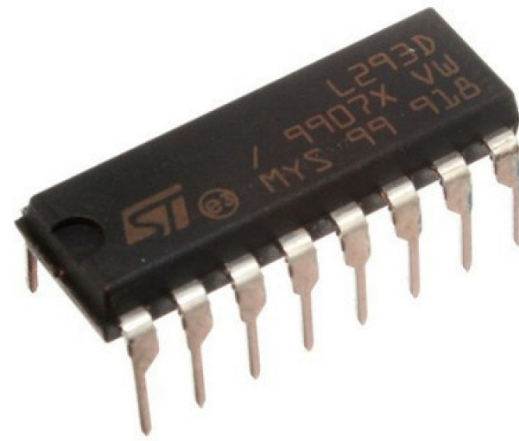


Motor Demo

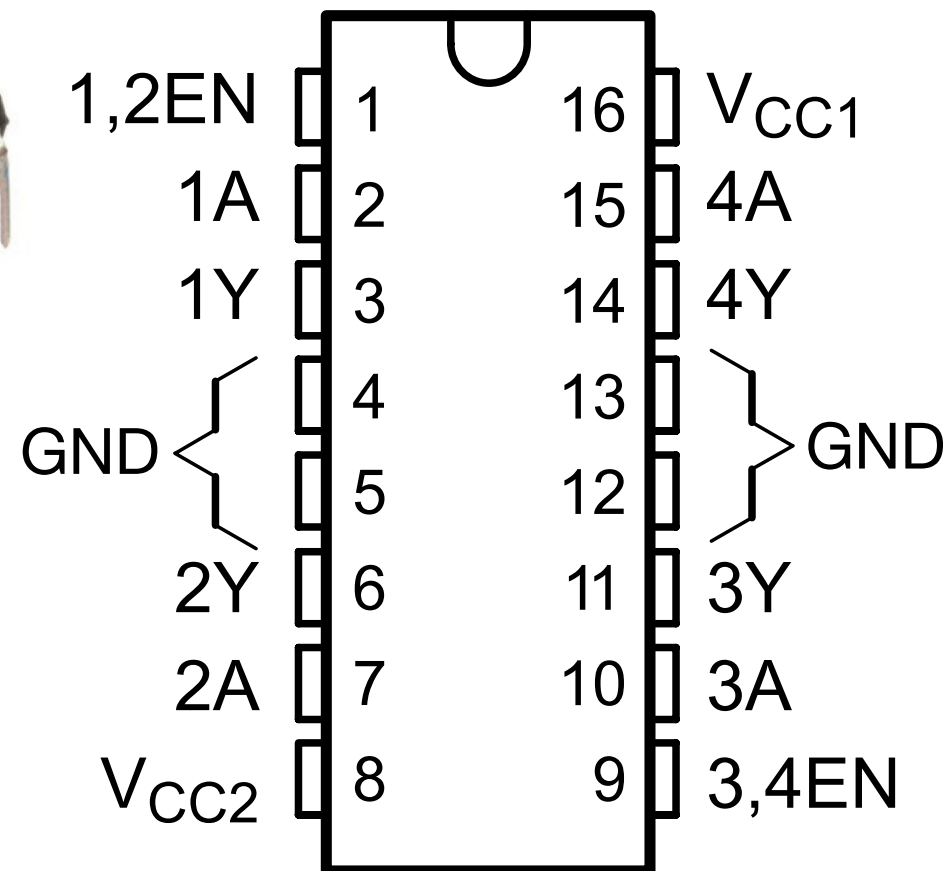
Motor



Motor Driver

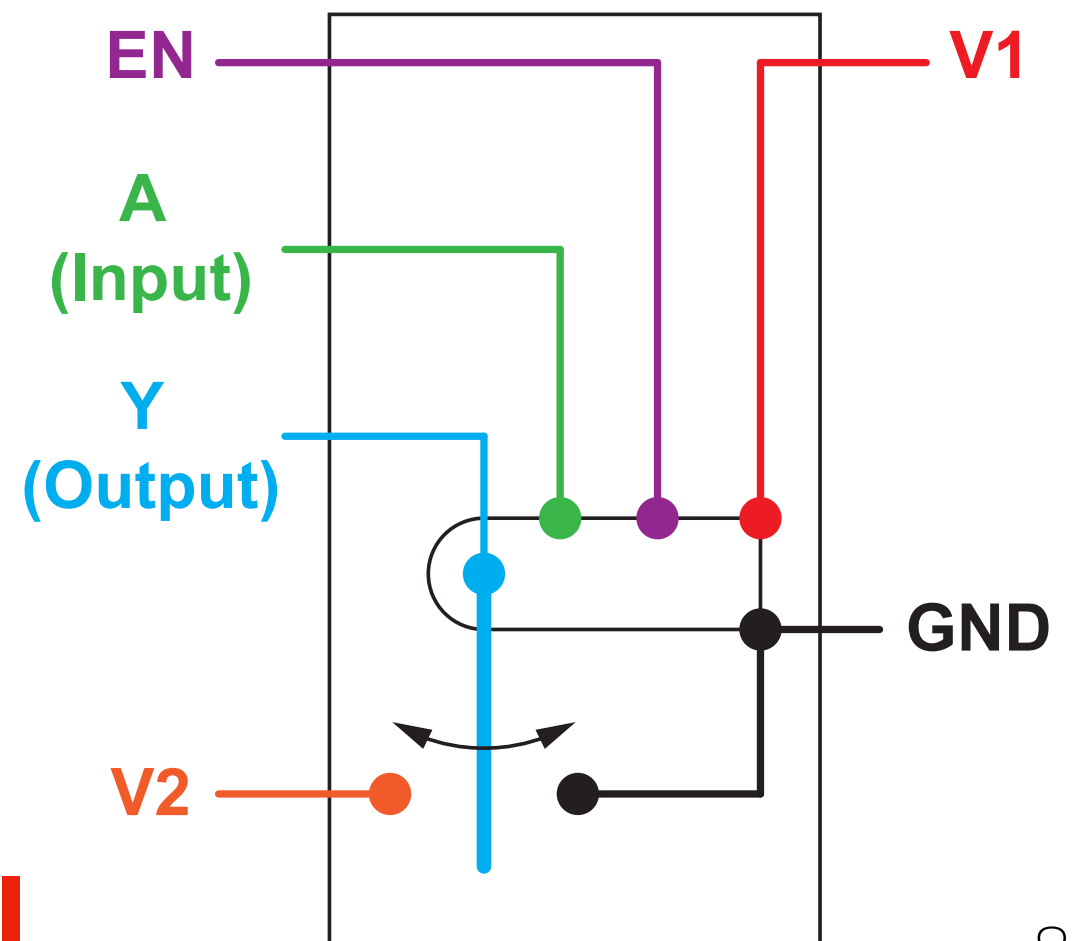


- Electronic switch
- Use low power signal to control high power circuit



```
if ((EN == V1) &&
    (A == V1)) {
    Y = V2;
} else {
    Y = GND;
}
```

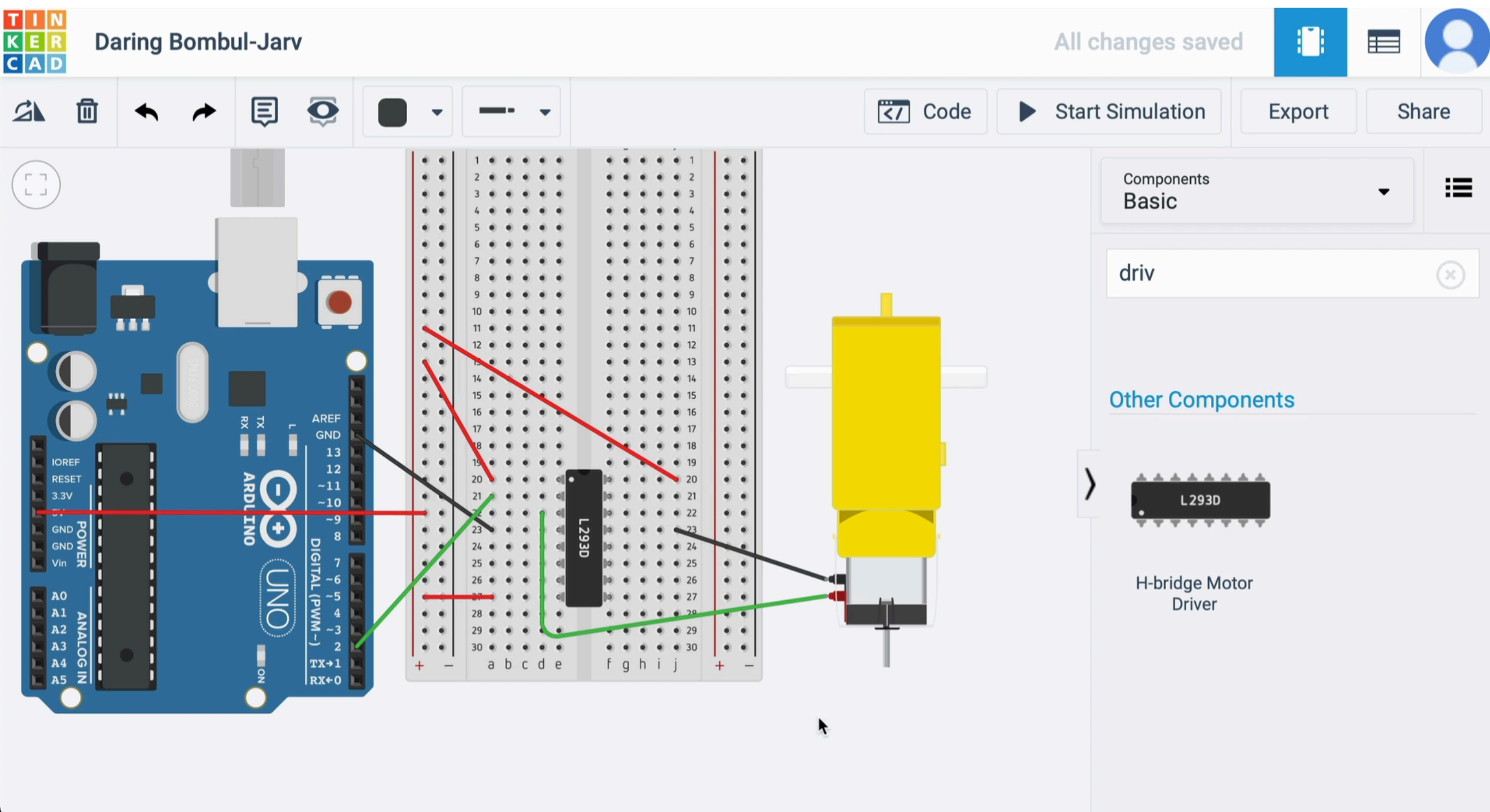
600 mA
Per channel



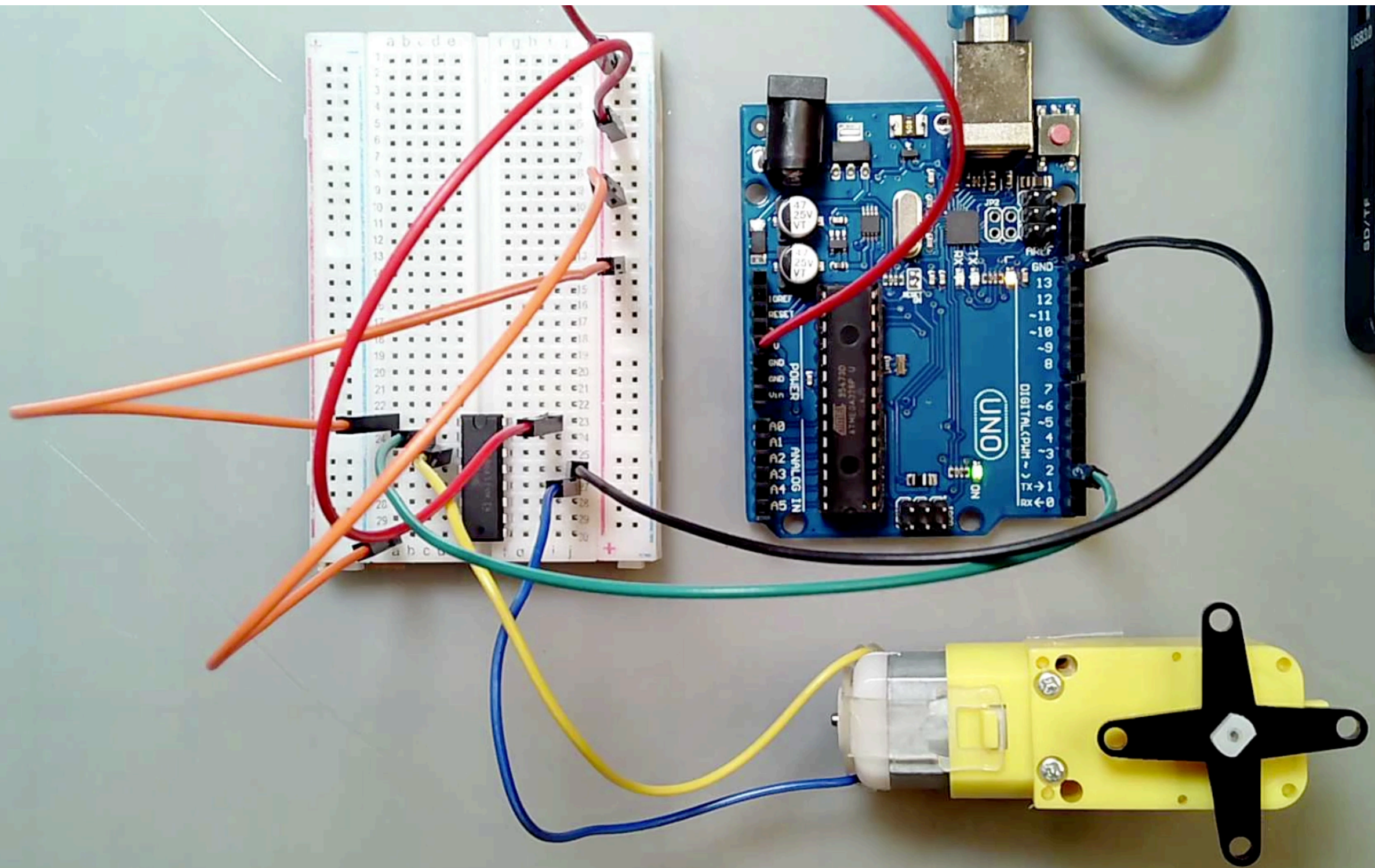
Motor Driver Demo

```
void setup() {  
    pinMode(2, OUTPUT);  
}  
  
void loop() {  
    digitalWrite(2, HIGH);  
    delay(1000);  
    digitalWrite(2, LOW);  
    delay(1000);  
}
```

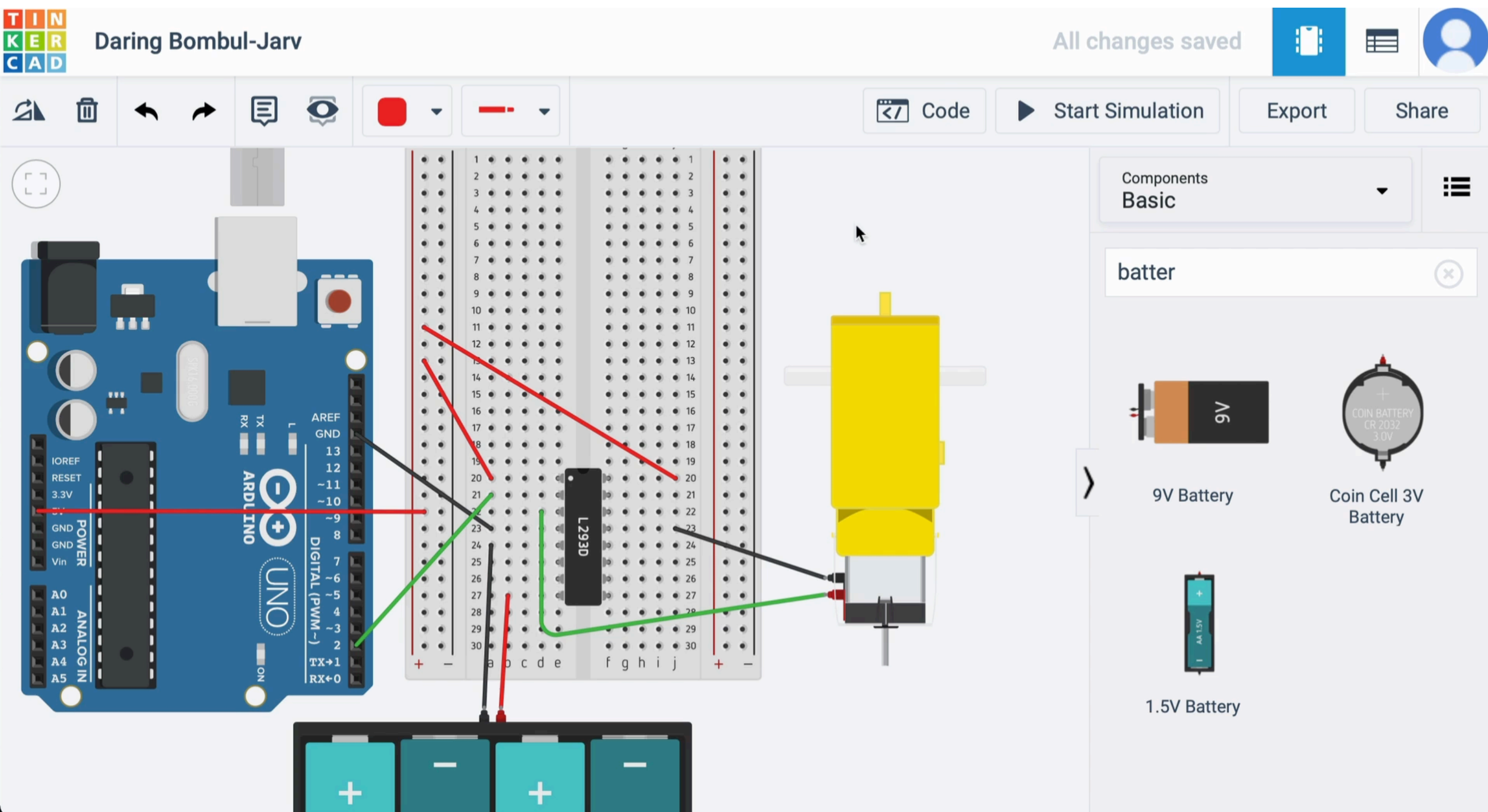
Motor Driver Demo



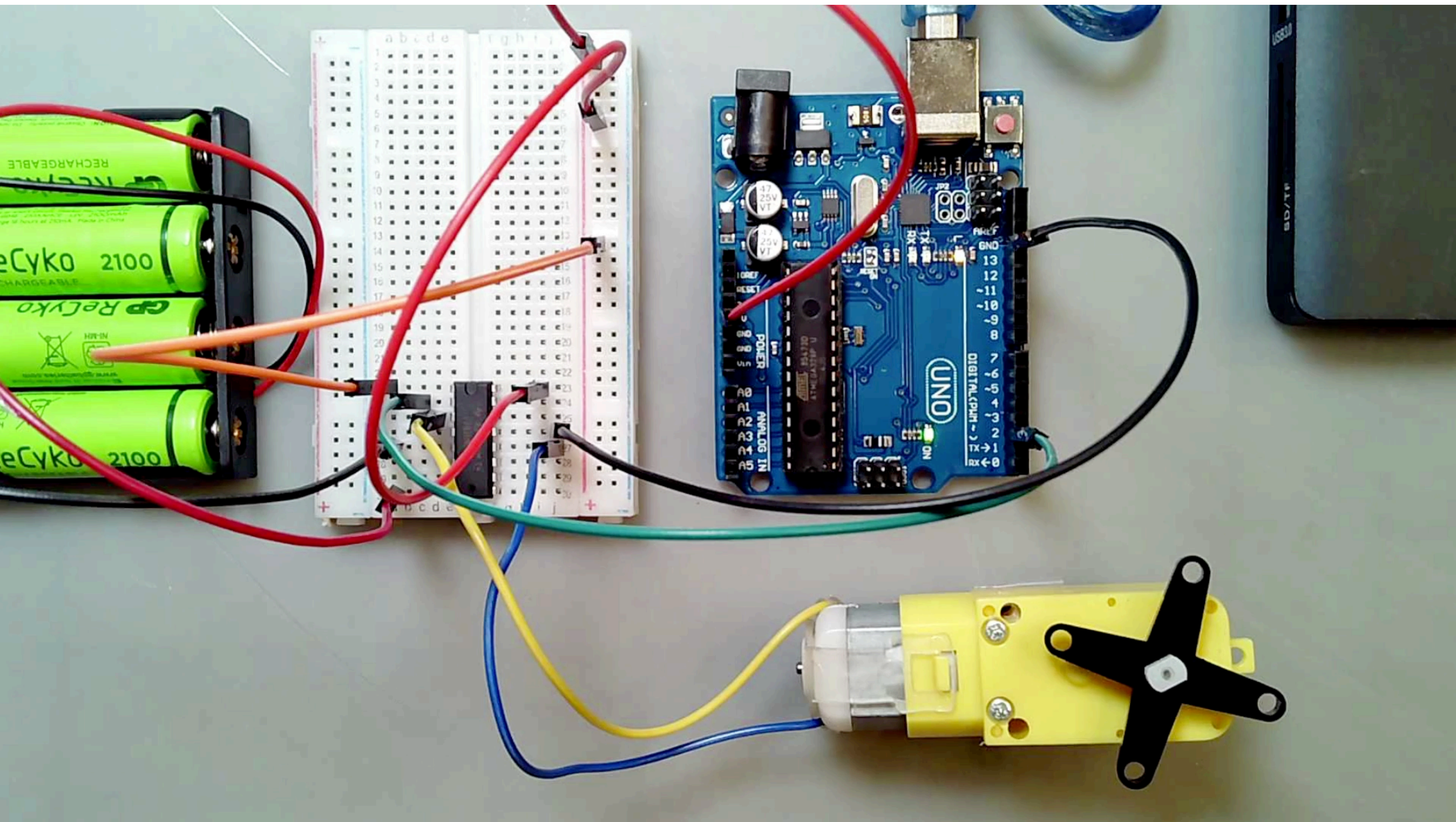
Motor Driver Demo



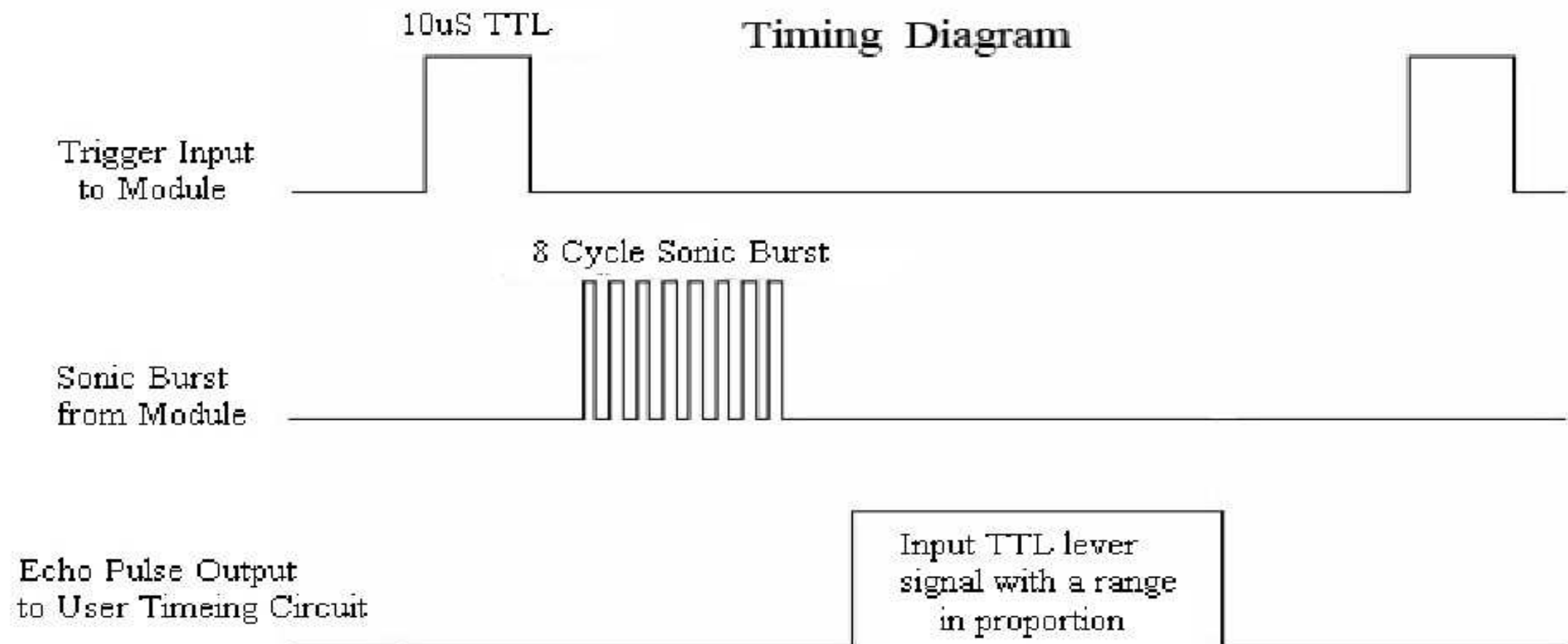
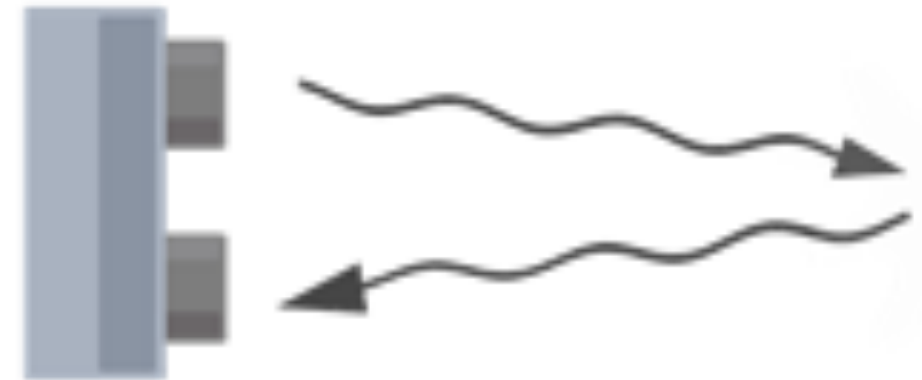
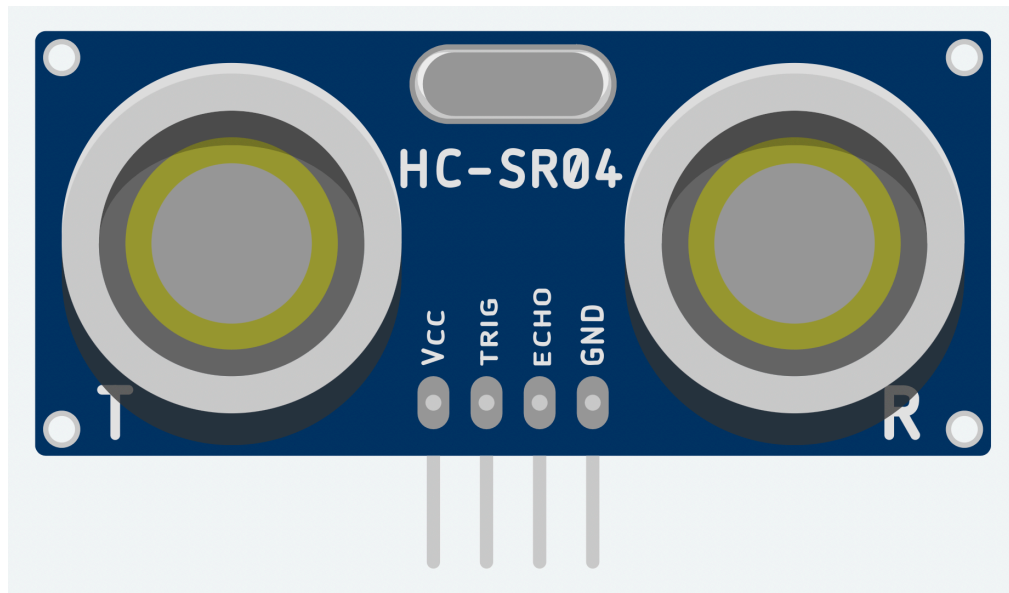
Motor Driver Demo



Motor Driver Demo



Ultrasonic Sensor



microseconds
`unsigned long pulseIn(pin, HIGH);`

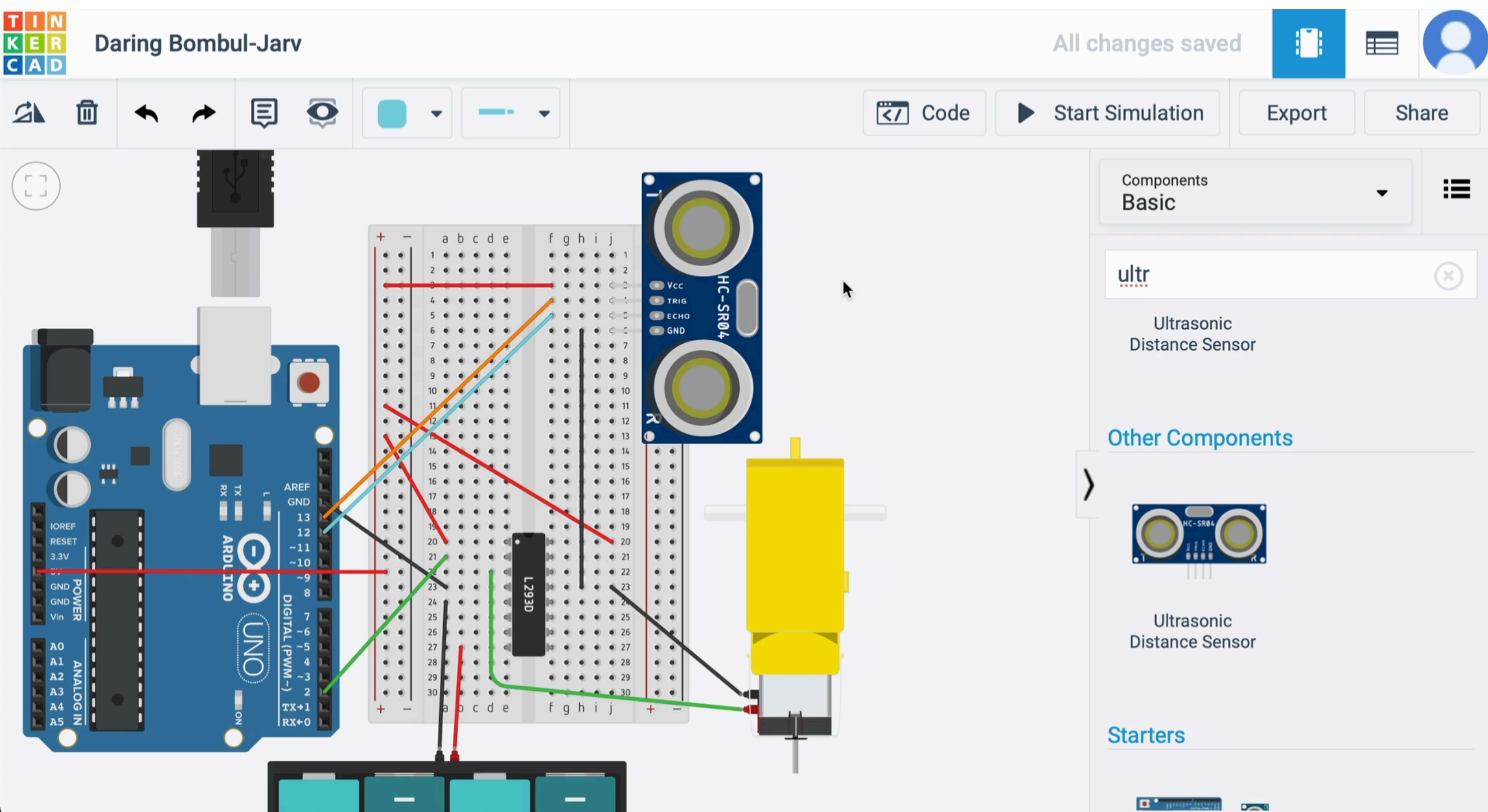
Sensor Demo

```
void setup() {  
    pinMode(motor_pin, OUTPUT);  
    pinMode(echo_pin, INPUT);  
    pinMode(trig_pin, OUTPUT);  
    digitalWrite(trig_pin, LOW);  
    Serial.begin(9600);  
}
```

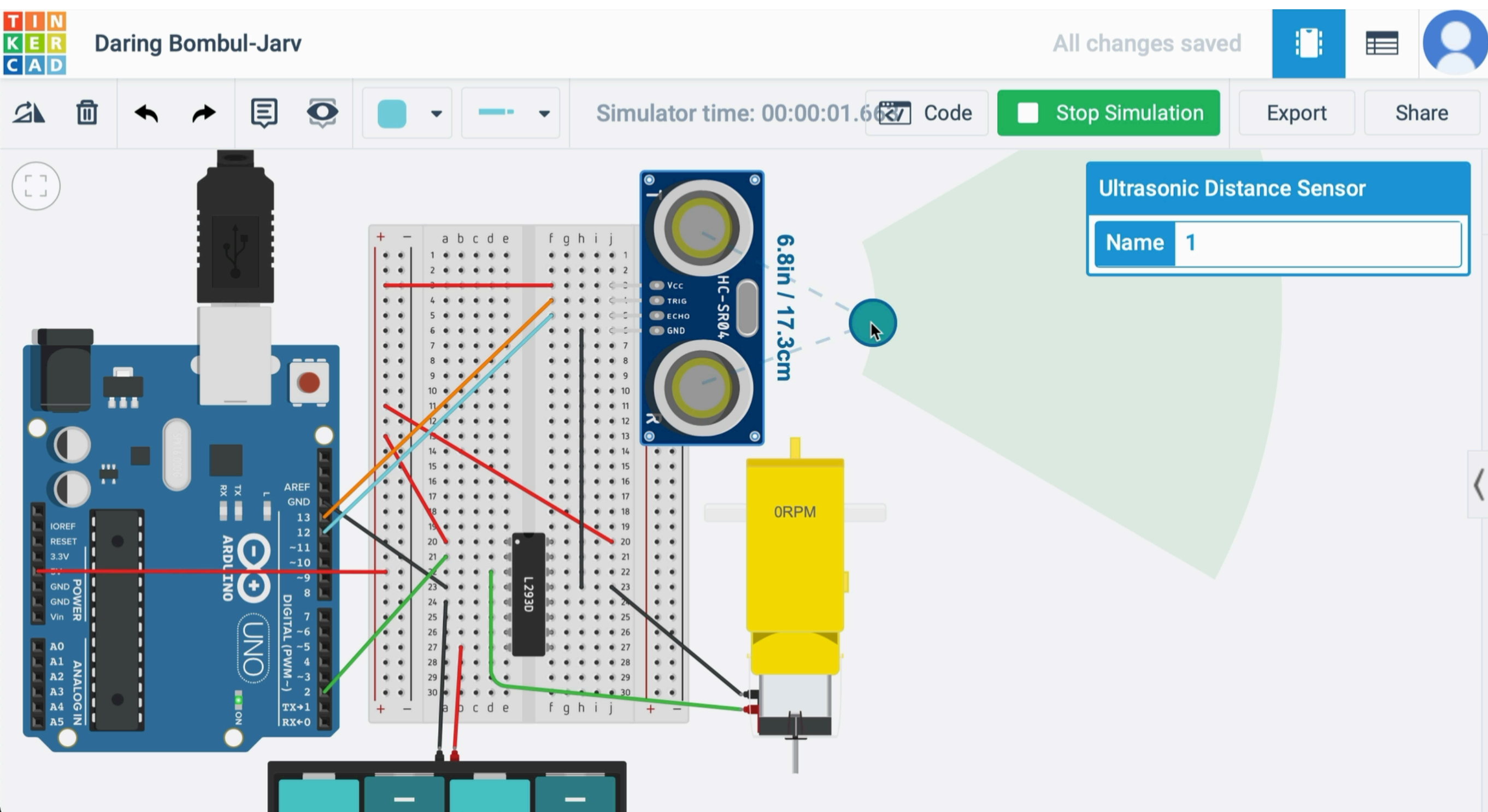
```
int motor_pin = 2;  
int trig_pin = 13;  
int echo_pin = 12;  
float sound_speed = 0.0345;  
// given in cm/microsec
```

```
void loop() {  
    digitalWrite(trig_pin, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(trig_pin, LOW);  
    int microsecs = pulseIn(echo_pin, HIGH);  
    float cms = microsecs*sound_speed/2;  
    Serial.println(cms);  
    if (cms < 30) {  
        digitalWrite(motor_pin, LOW);  
    } else {  
        digitalWrite(motor_pin, HIGH);  
    }  
}
```

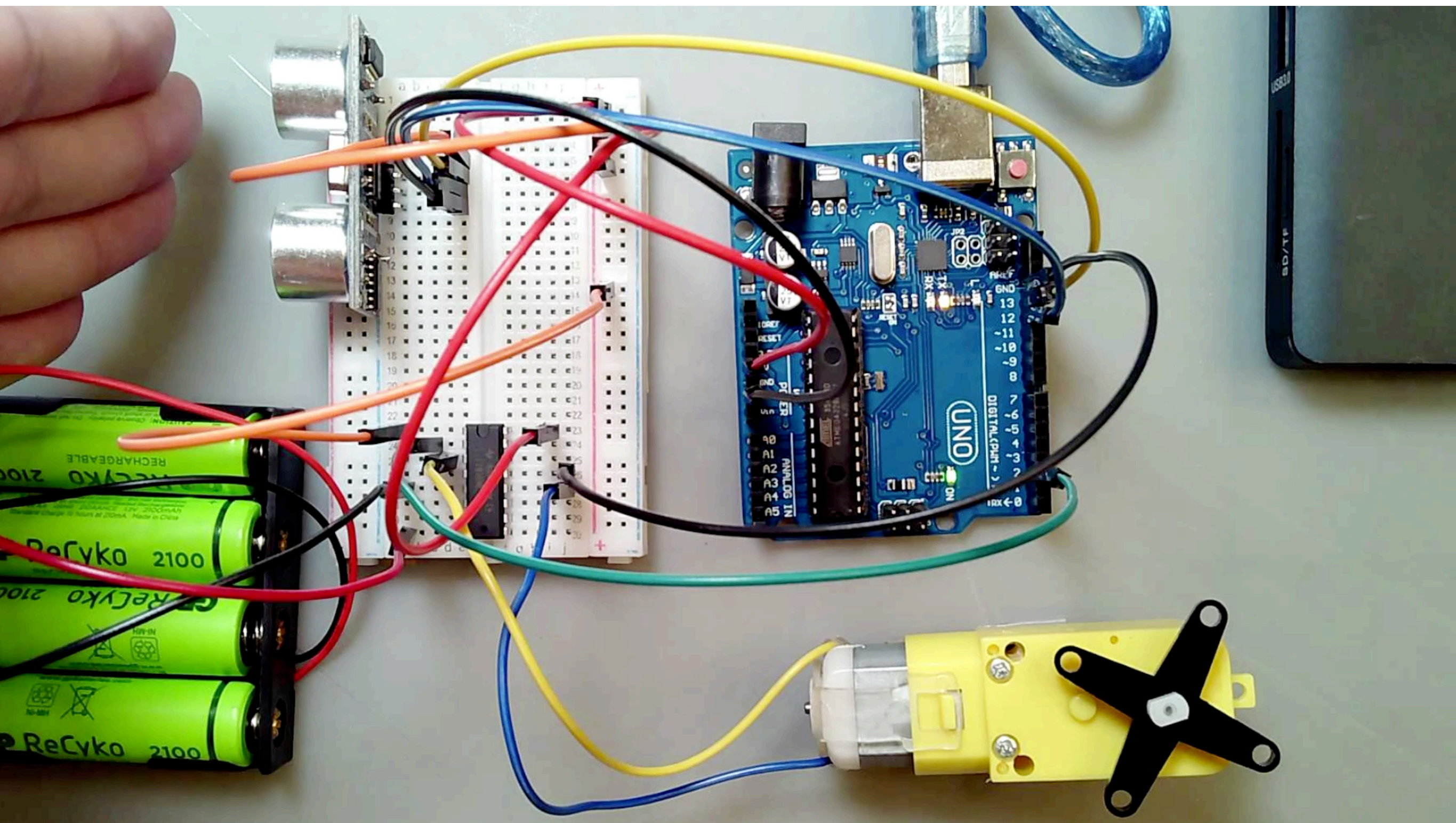
Sensor Demo



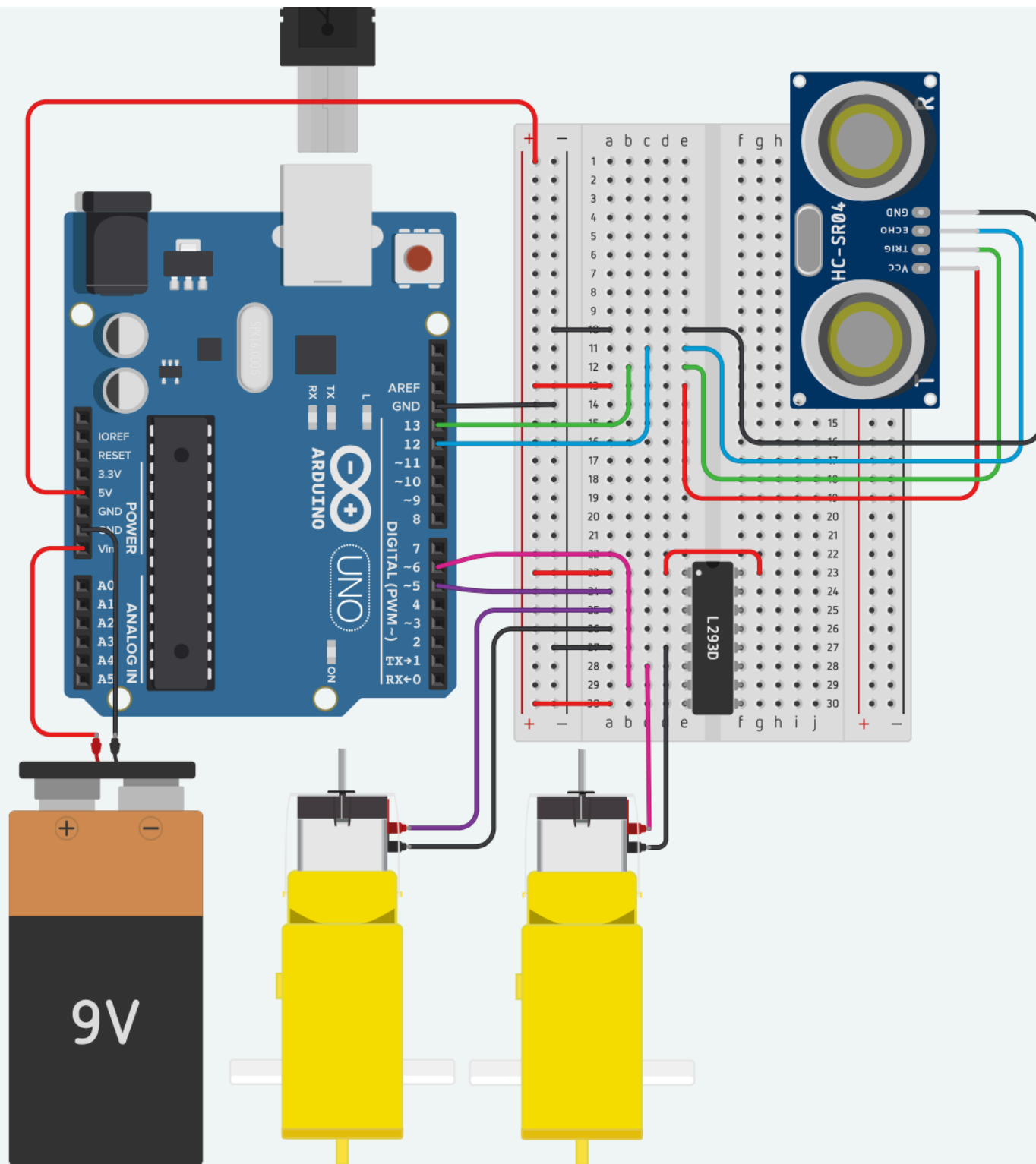
Sensor Demo



Sensor Demo



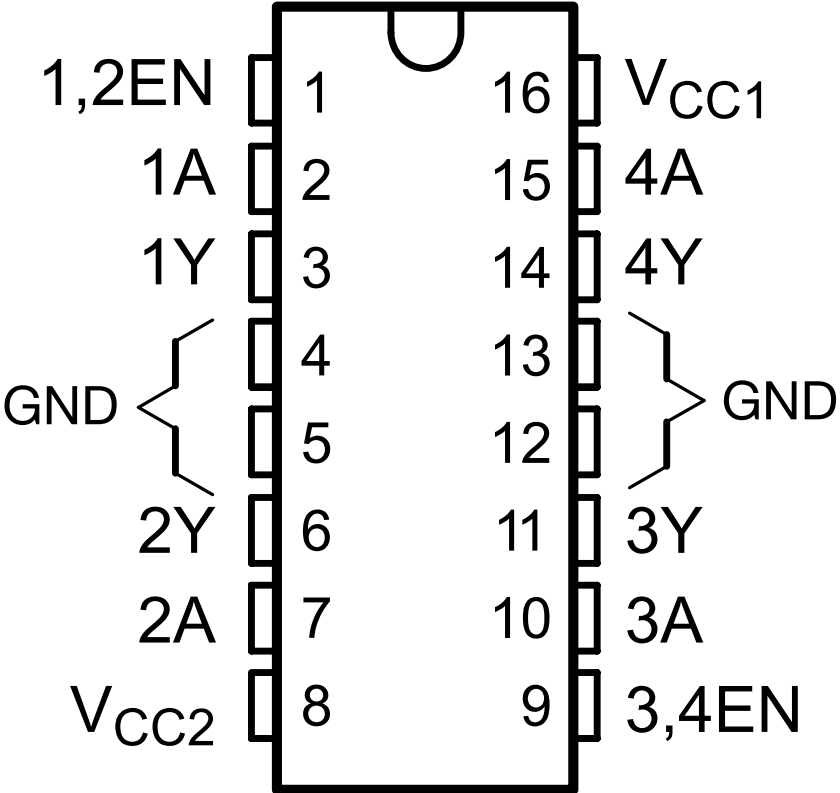
Template Project



```
1  int TRIG_PIN = 13;
2  int ECHO_PIN = 12;
3  int MOTOR_PIN1 = 6;
4  int MOTOR_PIN2 = 5;
5  float SPEED_OF_SOUND = 0.0345;
6
7  void setup() {
8      pinMode(MOTOR_PIN1, OUTPUT);
9      pinMode(MOTOR_PIN2, OUTPUT);
10     pinMode(TRIG_PIN, OUTPUT);
11     digitalWrite(TRIG_PIN, LOW);
12     pinMode(ECHO_PIN, INPUT);
13     Serial.begin(9600);
14 }
15
16 void loop() {
17     digitalWrite(TRIG_PIN, HIGH);
18     delayMicroseconds(10);
19     digitalWrite(TRIG_PIN, LOW);
20     int microsecs = pulseIn(ECHO_PIN, HIGH);
21     float cms = microsecs*SPEED_OF_SOUND/2;
22     Serial.println(cms);
23     if (cms < 30) {
24         digitalWrite(MOTOR_PIN1, LOW);
25         digitalWrite(MOTOR_PIN2, LOW);
26     } else {
27         digitalWrite(MOTOR_PIN1, HIGH);
28         digitalWrite(MOTOR_PIN2, HIGH);
29     }
30     delay(10);
31 }
```

Practice

On the L293D motor driver, if the GND pin is connected to ground, pins VCC1, 1,2EN, 2A to 5V, and VCC2 to 9V, approximately what voltage will be output on pin 2Y?



Respond at pollev.com/jku

A

B

C

D

- A) -5 V
- B) 0 V
- C) 5 V
- D) 9 V

Practice

When reading a signal from the HC-SR04 Ultrasonic Sensor, the `pulseIn` function will return an integer representing an amount of time measured in:

Respond at pollev.com/jku

A

B

C

D

- A) nanoseconds
- B) microseconds
- C) milliseconds
- D) seconds

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